

Test Colors (BGR)

Blue Color Detection (mask)

Blue Pixels Only

checkerboard.png

color_splash.png

composite_masterpiece.png

create_challenge_images.py

create_test_image_day3.py

create_test_image.py

crop.py

custom_cyberpunk.png

custom_mask.png

custom_sketch.png

custom_vintage.png

day5_transformation.py

day6_arithmetic_bitwise.py

day7_mini_challenge.py

day8_colour_spaces.py

dissolve.mp4

drawin_basics.py

drawing.py

ex_1_day_3.py

ex_2_day_3.py

OUTLINE

TIMELINE

345 print("-" * 70)

346 cv2.destroyAllWindows()

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Bright red BGR: (0,0,255) → HSV Hue: 0

Dark red BGR: (0,0,100) → HSV Hue: 0

Notice: Hue values are similar! This is why HSV i

[Section 7] Practical: Color Detection with HSV

Color detection with HSV is powerful for tracking

[Section 8] Merging channels back together...

Method 1: cv2.merge()

Open Quick Access

day7_mini_challenge.py

day8_colour_spaces.py

CHAT

Original (BGR)

BGR (OpenCV default)

RGB (Human perception)

HSV (Hue-Saturation-Value)

Grayscale

LAB (L*a*b*)

Grayscale - Standard (weighted)

Value Channel (brightness)

Saturation Channel (color purity)

Hue Channel (color information)

Red Channel Visualized (Red only)

Green Channel Visualized (Green only)

Blue Channel Visualized (Blue only)

Red Channel

Green Channel

Blue Channel (0 = no blue, 255 = max blue)

Merged back (should match original)

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11-06-2026

File Edit Selection View Go Run Terminal Help Python_Test

EXPLORER

- PYTHON_TEST
 - alignment_result.png
 - annotated.png
 - arithmetic.py
 - art_cyanotype.png
 - art_green screen.png
 - art_magenta dream.png
 - art_sepia.png
 - beach.png
 - before_after.png
 - bitwise.py
 - black.jpg
 - blended.png
 - challenge_blend.png
 - challenge_main.png
 - checkerboard.png
 - color_isolation_blue.png
 - color_isolation_green.png
 - color_isolation_purple.png
 - color_isolation_red.png
 - color_isolation_yellow.png
 - color_space_reference.png
 - color_splash.png
 - colorspaces.py
 - composite_masterpiece.png
 - create_challenge_images.py
 - create_test_image_day3.py
 - create_test_image.py
 - crop.py
 - custom_cyberpunk.png
 - custom_mask.png

exercise2_color_picker.py exercise3_color_isolation.py exercise4_color_s

colorspaces.py

```
15 # Load the image and show it
16 image = cv2.imread(args["image"])
17 cv2.imshow("Original", image)
18
19 # Convert the image to grayscale
20 gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
21 cv2.imshow("Gray", gray)
22
23 # Convert the image to the HSV (Hue, Saturation, Value)
24 # color spaces
25 hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
26 cv2.imshow("HSV", hsv)
27
28 # Convert the image to the L*a*b* color spaces
29 lab = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)
30 cv2.imshow("L*a*b*", lab)
31 cv2.waitKey(0)
```

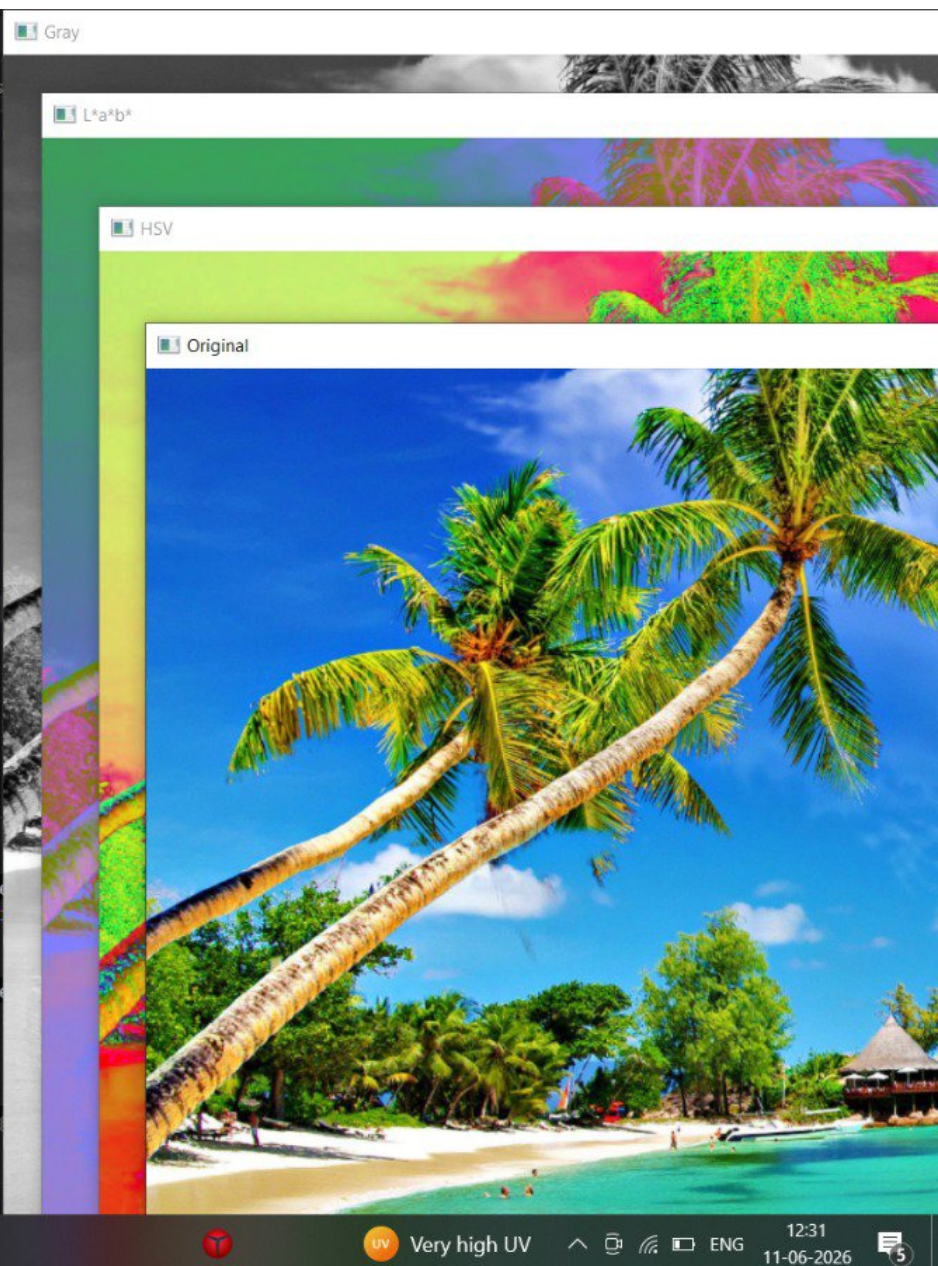
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

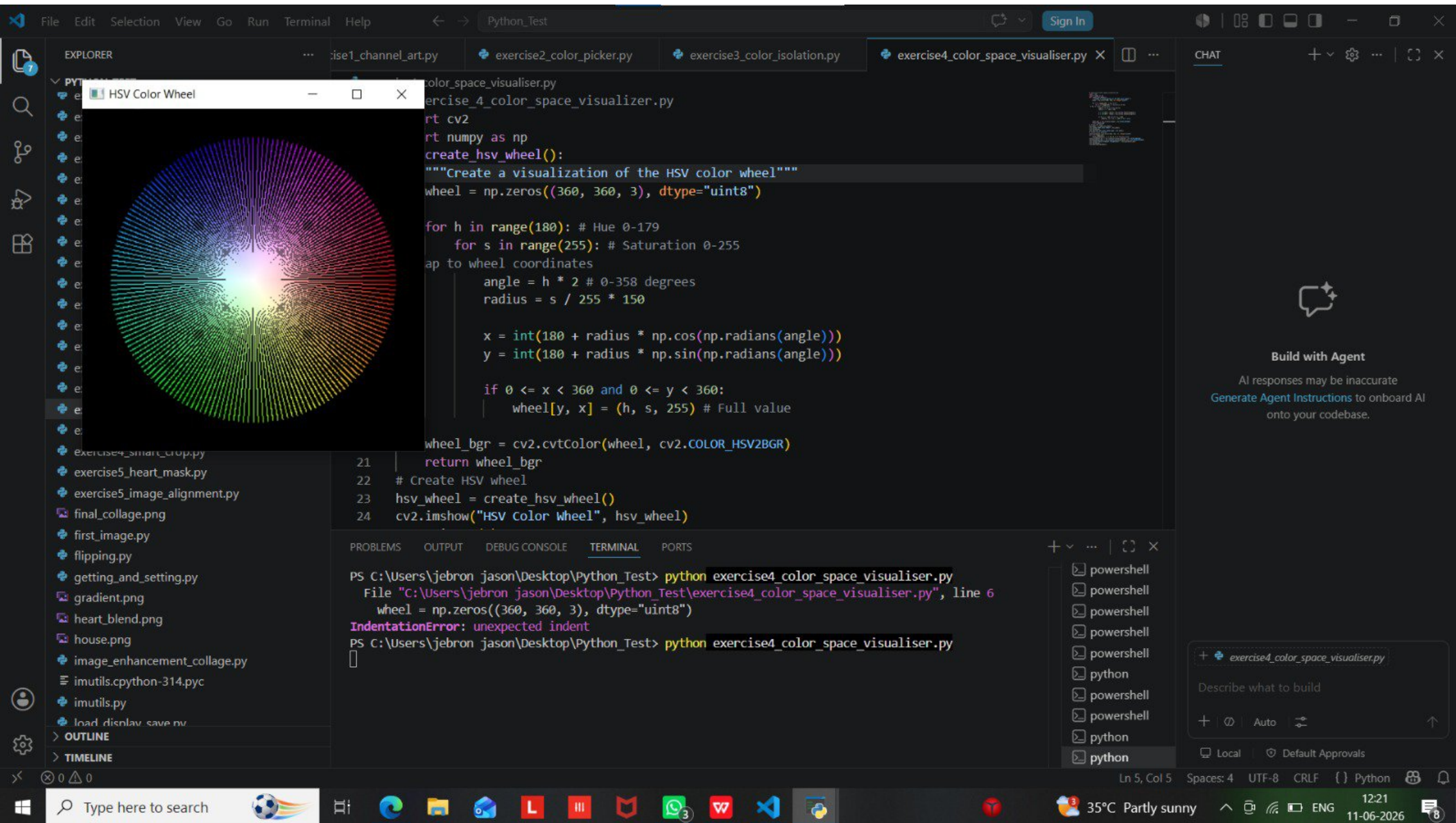
```
PS C:\Users\jeb_\jason\Desktop\Python_Test> python exercise4_color_space
File "C:\Users\jeb_\jason\Desktop\Python_Test\exercise4_color_space_v
wheel = np.zeros((360, 360, 3), dtype="uint8")
IndentationError: unexpected indent
PS C:\Users\jeb_\jason\Desktop\Python_Test> python exercise4_color_space
PS C:\Users\jeb_\jason\Desktop\Python_Test> python colorspace.py
usage: colorspace.py [-h] -i IMAGE
colorspaces.py: error: the following arguments are required: -i/--image
PS C:\Users\jeb_\jason\Desktop\Python_Test> python colorspace.py --ima
```

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Keep Only Red

Keep Only Purple

Keep Only Yellow

Keep Only Blue

Keep Only Green

Python_Test

8_colour_spaces.py exercise1_channel_art.py exercise2_color_picker.py exercise3_color_isolation.py

```
image = cv2.imread('...')
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
# Define ranges for different colors
color_ranges = {
    'red': ([0, 50, 50], [10, 255, 255]), ([160, 50, 50], [179, 255, 255]),
    'purple': ([40, 50, 50], [80, 255, 255]),
    'yellow': ([100, 50, 50], [130, 255, 255]),
    'blue': ([20, 50, 50], [30, 255, 255]),
    'green': ([130, 50, 50], [160, 255, 255])
}

for name, ranges in color_ranges.items():
    # Create a mask for this color
    mask = np.zeros(image.shape[:2], dtype="uint8")
    for lower, upper in ranges:
        lower = np.array(lower)
        upper = np.array(upper)
        mask = cv2.bitwise_or(mask, cv2.inRange(hsv, lower, upper))

# Isolate this color, others become grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
gray_3ch = cv2.cvtColor(gray, cv2.COLOR_GRAY2BGR)
result = np.where(mask[:, :, np.newaxis], image, gray_3ch)
```

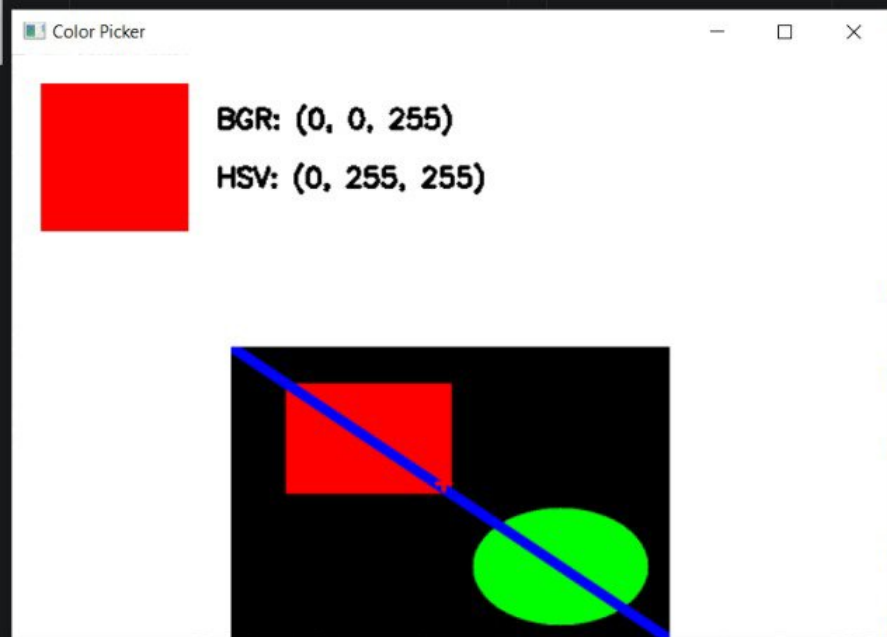
DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\jebbron\jason\Desktop\Python_Test> python exercise2_color_picker.py
Move mouse over the image to see color values
Press ESC to exit

PS C:\Users\jebbron\jason\Desktop\Python_Test> python exercise3_color_isolation.py
File "C:\Users\jebbron\jason\Desktop\Python_Test\exercise3_color_isolation.py", line 19
lower = np.array(lower)
^^^^
IndentationError: expected an indented block after 'for' statement on line 18

PS C:\Users\jebbron\jason\Desktop\Python_Test> python exercise3_color_isolation.py
PS C:\Users\jebbron\jason\Desktop\Python_Test> python exercise3_color_isolation.py

Windows taskbar: Type here to search, 35°C Partly sunny, 12:10 11-06-2026



```
ay8_colour_spaces.py exercise1_channel_art.py exercise2_color_picker.py
```

```
picker showing BGR and HSV values""
```

```
est_image.png")
```

```
t_image.png first!")
```

```
as
```

```
, 600, 3), dtype="uint8") * 255
```

```
ent, x, y, flags, param):
```

- exercise3_mask_tool.py
- exercise3_rotation_animation.py
- exercise4_colour_splash.py
- exercise4_smart_crop.py
- exercise5_heart_mask.py
- exercise5_image_alignment.py
- final_collage.png
- first_image.py
- flipping.py
- getting_and_setting.py
- gradient.png
- heart_blend.png
- house.png
- OUTLINE
- TIMELINE

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\jebroon\jason\Desktop\Python_Test> python exercise2_color_picker.py
Move mouse over the image to see color values
Press ESC to exit
```

+ ... | {} X

- powershell
- powershell
- powershell
- powershell
- powershell
- powershell
- python
- powershell
- powershell
- python

CHAT

Build with Agent

AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.

+ exercise2_color_picker.py

Describe what to build

+ | Auto

Local Default Approvals

Effect: Cyanotype

Python_Test

cement_collage.py day7_mini_challenge.py day8_colour_spaces.py exercise1_channel_art.py

```
effects = {
    'Cyanotype': cv2.merge([b, b, b]), # blue only
    'Sepia': cv2.merge([b, g, (r * 0.7).astype("uint8")]),
    'Magenta Dream': cv2.merge([
        (b * 0.5).astype("uint8"),
        (g * 0.3).astype("uint8"),
        r
    ]),
    'Green Screen': cv2.merge([
        (b * 0.2).astype("uint8"),
        g,
        (r * 0.2).astype("uint8")
    ]),
}
```

name, img in effects.items():
cv2.imshow(f"Effect: {name}", img)
cv2.waitKey(0)
cv2.imwrite(f"art_{name.lower()}.png", img)

destroyAllWindows()

Effect: Sepia

Effect: Green Screen

Effect: Magenta Dream

OUTPUT DEBUG CONSOLE TERMINAL PORTS

s\jebon jason\Desktop\Python_Test> python exercise1_channel_art.py
(most recent call last):
File "D:\a\opencv-python\opencv-python\opencv\modules\core\src\merge.dispatch.cpp", line 11, in <module>
 ta Dream": cv2.merge([b*0.5, g*0.3, r]),
 ~~~~~~  
OpenCV(4.13.0) D:\a\opencv-python\opencv-python\opencv\modules\core\src\merge.dispatch.cpp: (-215:Assertion failed) mv[i].size == mv[0].size && mv[i].depth() == depth in function '  
s\jebon jason\Desktop\Python\_Test> python exercise1\_channel\_art.py  
s\jebon jason\Desktop\Python\_Test> python exercise1\_channel\_art.py

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